# Thousands of AI Authors

What 2,778 researchers actually think the next 20 years look like

after Grace, Stewart, Sandkühler, Thomas, Weinstein-Raun, Brauner (2024)

# One survey. One question per expert.

**2,778**

AI researchers

every NeurIPS, ICML, ICLR, JMLR, AAAI, IJCAI author from the previous year

Asked in 2023.

Same questions as 2022.

Largest expert survey of its kind. Answers sealed before GPT-4 retail launch.

The story of this paper is four moments.

Each one makes you stop.

1. A thirteen-year shift.
2. A dot plot that already aged badly.
3. A tail that will not go away.
4. What they actually fear.

# Moment 1: The thirteen-year shift

# The question

*"When unaided machines can accomplish every task better and more cheaply than human workers."*

They call it **High-Level Machine Intelligence** (HLMI).
Call it AGI if you prefer.

Verbatim definition given to respondents, Grace et al. (2024), p. 4.

# In 2022, the aggregate answer was:

**2060**

50% probability date for HLMI

Roughly a generation away.

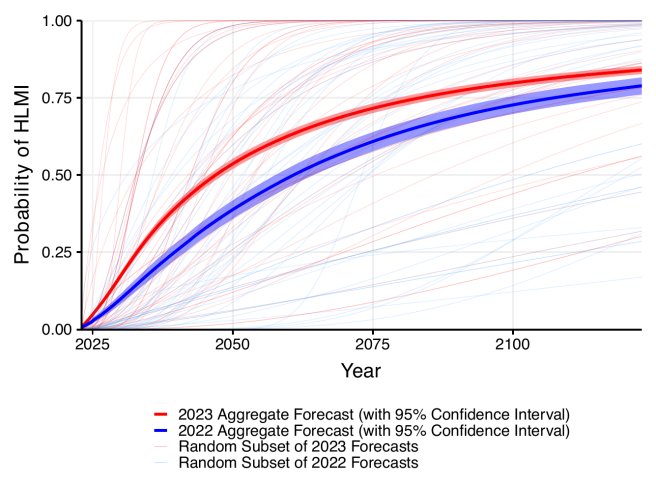
# One year later, the same experts said:

**2047**

50% probability date for HLMI

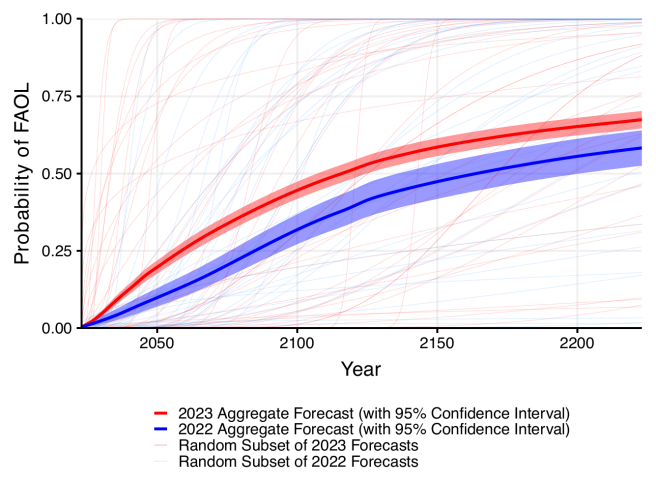
Thirteen years closer.

# The shift, as a picture




Aggregate forecast: 10%, 50%, and 90% probability dates for HLMI, 2022 vs. 2023.

# And for “all human jobs”




Full Automation of Labor (FAOL) also shifted, by 48 years.

# Read that again

The 50% date for automating
**every job on earth**
moved from 2164 to 2116.

# Moment 2: A dot plot that already aged badly

# “HLMI” is too abstract.

So they asked about 39 specific things.

Write a *New York Times* best-seller.
Fold laundry.
Translate into a newfound language.
Do your taxes. Play Starcraft from screen video.

For each: when does 50% probability cross?

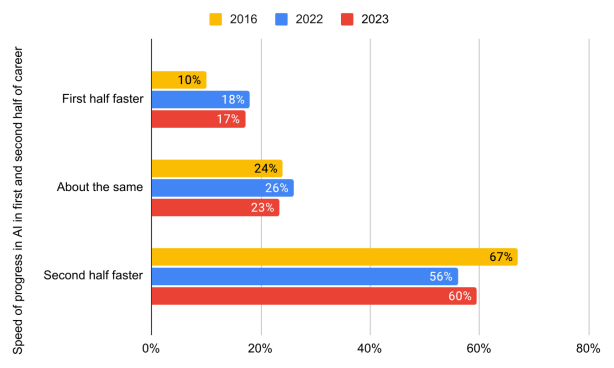
Task
Job
2022
2023
Year
2050
2100
2150
Full Automation of Labor (all human jobs)
AI Researcher
High Level Machine Intelligence (all human tasks)
Surgeon
Publishable math theorems
NYT best-selling fiction
Translate text in newfound language
Retail Salesperson
Win Putnam Math Competition
Simple Python code given spec and examples
Answers undecided questions well
Answer open-ended fact questions with web
Explain AI actions in games
5km city race as bipedal robot (superhuman)
Equations governing virtual worlds
Top 40 Pop Song
Translate text (vs. fluent amateur)
Transcribe speech
Answer factoid questions with web
Beat humans at Go (after same # games)
Recognize object seen once
Fake new song by specific artist
Translate speech using subtitles
Telephone banking services
Read text aloud
Random new computer game (novice level)
Construct video from new angle
Group new objects into classes
Good high school history essay
World Series of Poker
Atari games after 20m play (50% vs. novice)
Angry Birds (superhuman)
Learn to sort long lists
All Atari games (vs. pro game tester)
Fold laundry
Truck Driver
Assemble LEGO given instructions
Top Starcraft play via video of screen
Milestones
2025
2030
2035
Year

# Look at the top of that chart

- **Write a NYT best-seller.** Forecast: 2033.
- **Answer open-ended factual questions.** Forecast: 2027.
- **Write simple Python from a spec.** Forecast: 2025.

Several of these were arguably done before the ink dried.

# Pace of progress: then vs. now




Respondents were asked to compare the **speed of AI progress** in the *first half* vs. *second half* of their own time working in the field.

"Career" here = years spent working in AI; median ≈ 5 years.

A clear majority say progress is **faster now** than when they started.

Grace et al. (2024), Figure 4.

# Moment 3: A tail that will not go away

# They were also asked about outcomes.

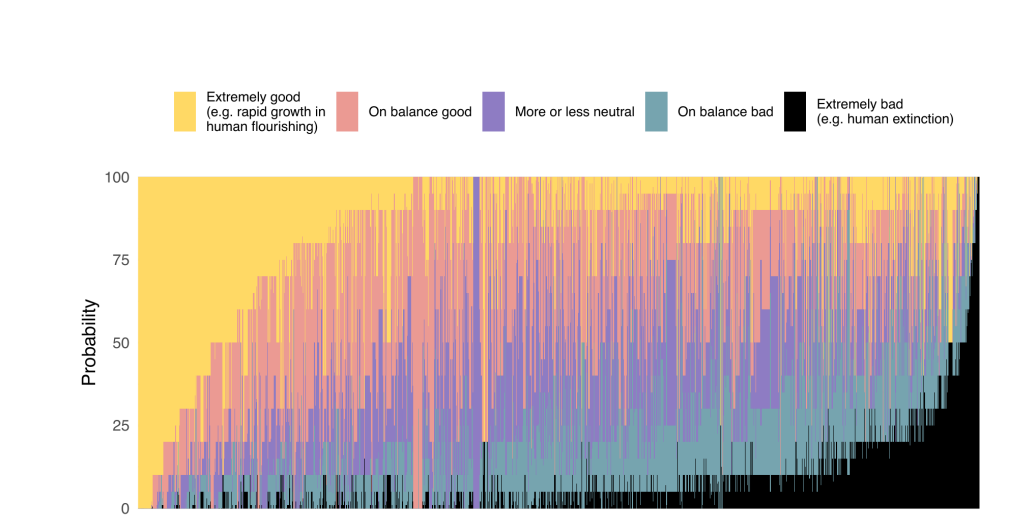
The exact question asked:

*"Assume for the purpose of this question that HLMI will at some point exist. How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"*

Respondents assigned probabilities (summing to 100%) across five buckets:

extremely good · on balance good · neutral · on balance bad · extremely bad (e.g. human extinction)

Verbatim, Grace et al. (2024), footnote 5, p. 14; see also §4.2, p. 12.




Each vertical bar is one respondent, sorted by optimism.

# The number that stops a boardroom

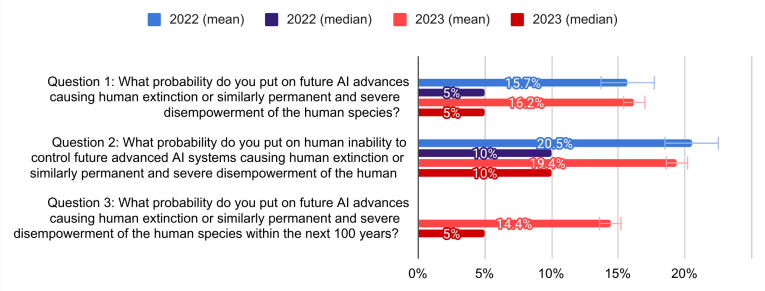
**38%**

of surveyed AI researchers
give at least 10% probability
to outcomes *"as bad as human extinction"*.

Quoted phrase: Grace et al. (2024), abstract, p. 1 and §4.2, p. 12.

The paper's range across framings is 37.8%–51.4%.

# The extinction-tail distribution




Respondent-level probability assigned to extinction-grade outcomes.

# And “intelligence explosion”

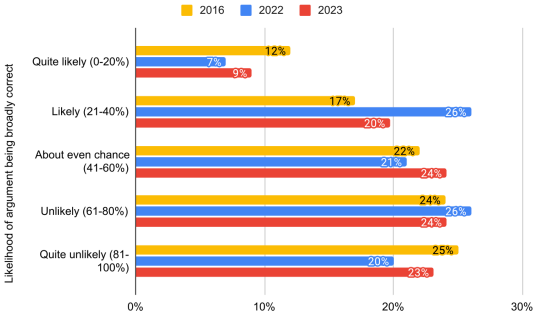



Figure 6: **Since 2016 a majority of respondents have thought that it’s either “quite likely,” “likely,” or an “about even chance” that technological progress becomes more than an order of magnitude faster within 5 years of HLMI being achieved.**

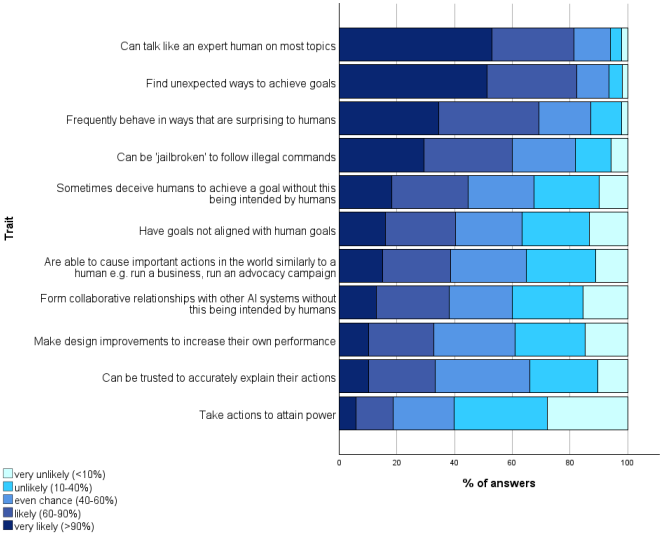
Likelihood that AI improves AI in a rapid, self-reinforcing loop.

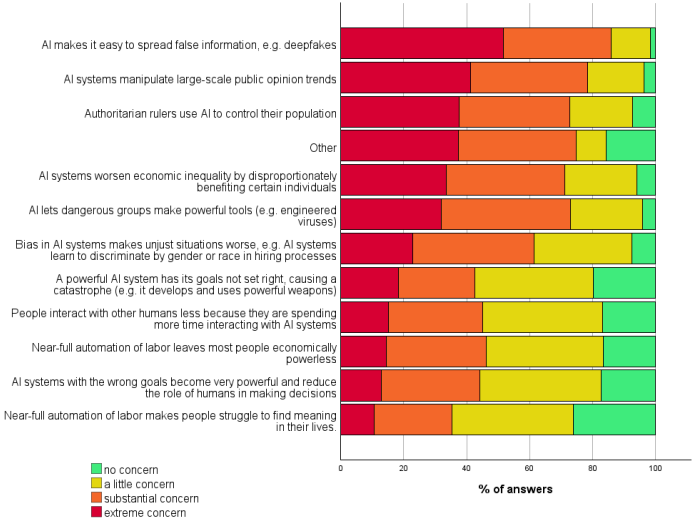
# Moment 4: What they actually fear

# Not robots. Not sentience.

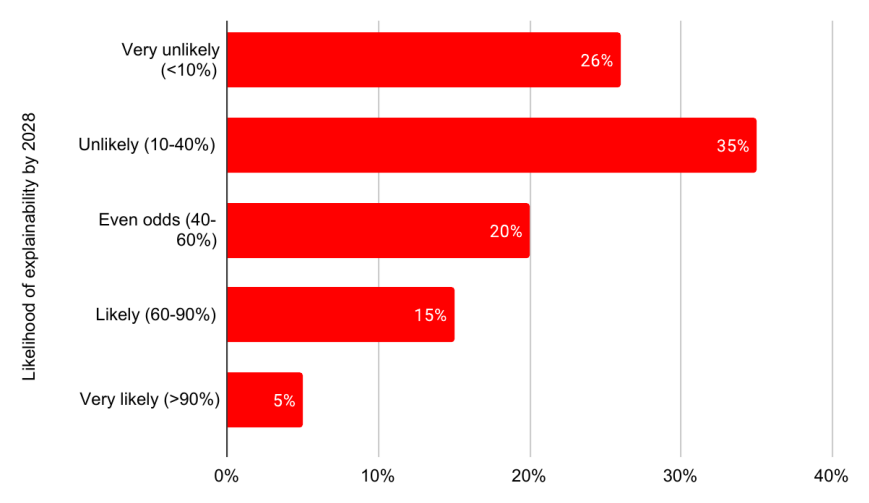
Two behaviours keep coming up.

**Deception.**
**Power-seeking.**


Trait
Can talk like an expert human on most topics
Find unexpected ways to achieve goals
Frequently behave in ways that are surprising to humans
Can be 'jailbroken' to follow illegal commands
Sometimes deceive humans to achieve a goal without this being intended by humans
Have goals not aligned with human goals
Are able to cause important actions in the world similarly to a human e.g. run a business, run an advocacy campaign
Form collaborative relationships with other AI systems without this being intended by humans
Make design improvements to increase their own performance
Can be trusted to accurately explain their actions
Take actions to attain power
0
20
40
60
80
100
% of answers
very unlikely (<10%)
unlikely (10-40%)
even chance (40-60%)
likely (60-90%)
very likely (>90%)


AI makes it easy to spread false information, e.g. deepfakes
AI systems manipulate large-scale public opinion trends
Authoritarian rulers use AI to control their population
Other
AI systems worsen economic inequality by disproportionately benefiting certain individuals
AI lets dangerous groups make powerful tools (e.g. engineered viruses)
Bias in AI systems makes unjust situations worse, e.g. AI systems learn to discriminate by gender or race in hiring processes
A powerful AI system has its goals not set right, causing a catastrophe (e.g. it develops and uses powerful weapons)
People interact with other humans less because they are spending more time interacting with AI systems
Near-full automation of labor leaves most people economically powerless
AI systems with the wrong goals become very powerful and reduce the role of humans in making decisions
Near-full automation of labor makes people struggle to find meaning in their lives.
0
20
40
60
80
100
% of answers
no concern
a little concern
substantial concern
extreme concern

# And what they want spent on it




Forecast likelihood that AI systems can truthfully and intelligibly explain their decisions

# So what?

# Three things to leave this room with

1. Expert timelines are **unstable**.
   Do not anchor a five-year plan on a point estimate.

2. Narrow-task forecasts **already under-shot**.
   Assume your workflow-level dot arrives early, not late.

3. The tail is **not a fringe**.
   38% of the field assigns $\geq$10% to catastrophic outcomes.

# The one number

**13**

years of expected progress
compressed into *twelve months* of calendar time.

Plan as if that could happen again.

# Thank you.

Questions.

Grace, K. *et al.* (2024).
Thousands of AI Authors on the Future of AI.
arXiv:2401.02843.

# Every quoted phrase, with page number i

All quotations in this deck, verified against *Grace et al. (2024), "Thousands of AI Authors on the Future of AI"*, arXiv:2401.02843.

- **HLMI definition:** "unaided machines can accomplish every task better and more cheaply than human workers."
  *Source:* p. 4, verbatim from the survey instrument.
- **Outcomes question:** "Assume for the purpose of this question that HLMI will at some point exist. How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"
  *Source:* footnote 5, p. 14; summarised in §4.2, p. 12.

# Every quoted phrase, with page number ii

- **Five outcome buckets:** "extremely good . . . on balance good . . . more or less neutral . . . on balance bad . . . extremely bad (e.g. human extinction)."
  *Source:* footnote 5, p. 14.
- **"As bad as human extinction":** "Between 37.8% and 51.4% of respondents gave at least a 10% chance to advanced AI leading to outcomes as bad as human extinction."
  *Source:* abstract, p. 1; figure and discussion in §4.2, p. 12.
- **HLMI aggregate dates:** 2060 (2022 survey) → 2047 (2023 survey); 13-year shift.
  *Source:* §4.1, Figure 3a.
- **FAOL aggregate dates:** 2164 (2022) → 2116 (2023); 48-year shift.
  *Source:* §4.1, Figure 3b.